

Ndayeni Solutions — Project Document

Development Team

Ndayeni Solutions Website — Project Document

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Author: Development Team

Repository: <https://github.com/ndayenisolutions-png/Ndayeni-Solutions>

Live URL: <https://www.ndayenisolutions.co.za>

Admin URL: <https://www.ndayenisolutions.co.za/training/admin>

1. Project Overview

Ndayeni Solutions is a Midrand-based South African IT services company founded in 2023 by Nhlakanipho Ntshangase. This project is the company's public website + two backend management systems:

1. **Marketing Website** — public-facing site with 9 service pages, vacancies/careers page, contact form, FAQ, legal pages (Privacy Policy + Terms of Service), and SEO infrastructure
2. **Student Management System (SMS)** — a digital academy admin for managing students, courses, attendance, assessments, certificates, and reports
3. **Vacancies Admin** — a standalone careers management system for publishing job postings

The site serves small businesses and homes across South Africa with services including IT support, computer repairs, networking, CCTV, web design, graphic design, digital automation, and digital skills training.

2. Technology Stack

Core Framework

Technology	Version	Purpose
Next.js	16.1.1+	React framework with App Router, Server Components, API routes
React	19.0.0+	UI library
TypeScript	5.x	Type-safe JavaScript (strict mode enabled)
Tailwind CSS	4.x	Utility-first CSS framework
shadcn/ui	New York style	Component library built on Radix UI primitives

Backend & Database

Technology	Version	Purpose
Prisma ORM	6.11.1+	Database ORM with PostgreSQL (Supabase)
Supabase	—	PostgreSQL database hosting (pooler + direct connection)
nodemailer	9.1.1+	SMTP email sending (mail.ndayenisolutions.co.za:587)
pdfkit	0.20.2	Server-side PDF generation (welcome letters)
qrcode	1.5.4	QR code generation (certificate verification)

Frontend Libraries

Technology	Version	Purpose
GSAP	3.15.0+	Scroll-triggered animations (entrance animations on sections)
Recharts	2.15.4+	Charts (reports dashboard: line, bar, pie charts)
lucide-react	0.525.0+	Icon library
framer-motion	12.23.2	<i>(Installed but no longer imported — removed from bundle via tree-shaking)</i>

Infrastructure

Technology	Purpose
Vercel	Hosting + auto-deploy on GitHub push + edge CDN + Web Analytics
GitHub	Source control (private repo: ndayenisolutions-png/Ndayeni-Solutions)
Afrihost	DNS management (ndayenisolutions.co.za domain)
Mail server	mail.ndayenisolutions.co.za:587 (SMTP with STARTTLS)

Development Tools

Tool	Purpose
Bun	Package manager + JS runtime
ESLint	Code linting
Turbopack	Next.js dev server bundler

3. Architecture

Application Structure

```
src/
├── app/                                # Next.js App Router
│   ├── page.tsx                       # Homepage (client component)
│   ├── layout.tsx                     # Root layout (metadata, fonts, JSON-LD,
│   │   │   Toaster, Analytics)
│   ├── globals.css                   # Global styles + Tailwind + design tokens
│   ├── sitemap.ts                    # Dynamic sitemap (16 entries)
│   ├── privacy-policy/               # POPIA privacy policy page
│   ├── terms/                        # Terms of Service page
│   ├── services/                    # 9 service landing pages + index
│   ├── vacancies/                   # Public vacancies page + detail page + admin
│   └── training/                    # Academy: landing, apply, admin (SMS),
│       certificate, verify, forgot-password
├── api/                               # API routes
│   ├── academy/                     # SMS API (17 route files)
│   └── contact/                     # Contact form API
├── components/
│   ├── ndayeni/                     # Public site components (14 files)
│   ├── academy/                     # Admin/SMS components (11 files)
│   └── ui/                           # shadcn/ui primitives (44 files)
├── lib/                               # Shared libraries
│   ├── academy-auth.ts              # Authentication (PBKDF2, HMAC sessions,
│   │   password validation)
│   ├── academy-session.ts           # Session extraction from cookies
│   ├── db.ts                        # Prisma client
│   ├── rate-limit.ts                # In-memory rate limiter
│   ├── welcome-letter.ts            # PDF generation logic
│   ├── section-images.ts            # Image path configuration
│   └── utils.ts                     # Utility functions (cn class merger)
└── hooks/
    └── use-toast.ts                 # Toast notification hook
```

Design System

- **Theme:** Dark mode (primary background: `#071515` / `bg-dark-deep`)
- **Brand colors:** Navy `#1e3a5f` (brand), Teal `#14b8a6` (accent/brand-light)
- **Typography:** Inter (sans-serif) + Space Grotesk (mono) via `next/font`
- **Glassmorphism:** `.glass` and `.glass-strong` utility classes (blur scoped to desktop only via `@media (hover: hover) and (pointer: fine)`)
- **Fonts on PDF:** Helvetica 11pt body, Helvetica-Bold 14pt headings, 9pt footer

4. Database Schema

Provider: Supabase (PostgreSQL)

ORM: Prisma 6.x

Schema file: `prisma/schema.prisma`

Models

Model	Description	Key Fields
User	Marketing site user (unused in production)	id, email, name
Post	Blog posts (unused)	id, title, content
AcademyUser	Admin/staff accounts	id, email, passwordHash (PBKDF2), name, role (super/admin/admissions/training/readonly), active, createdAt
Student	Student/applicant records	id, studentNumber, applicationRef, fullName, email, phone, idNumber, dateOfBirth, gender, nationality, address, selectedCourses, courseId, preferredStartDate, preferredMode, highestEducation, employmentStatus, nextOfKin*, status (applied → enrolled → completed), progress, notes, enrolledAt, completedAt, certificates, attendances, assessments, auditLogs
Course	Training courses	id, code (unique), title, description, duration, deliveryMethod, entryRequirements, fee, active, maxStudents, modules
Module	Course modules	id, courseId, title, description, duration, order, learningObjectives, active
Attendance	Attendance records	id, studentId, date, status (present/absent/excused), notes
Assessment	Assessment records	id, studentId, moduleTitle, date, result (pass/not-yet-competent), mark, comments
Certificate	Issued certificates	id, studentId, programName, studentName, idNumber, issueDate, certificateNumber (unique), signedBy, status (active/revoked/reissued)
AuditLog	Audit trail	id, userId, studentId, action, details, timestamp
Vacancy	Job postings	id, title, department, location, employmentType, description, responsibilities, requirements, benefits, closingDate, howToApply, status (draft/active), createdAt, updatedAt

Student Status Flow

applied → under-review → info-required → accepted → enrolled → active →
completed

- ↘ rejected
 - ↘ withdrawn
 - ↘ deferred
-

5. Routes & Pages

Public Pages (20 routes)

Route	Type	Description
/	Client	Homepage with 12 sections (Hero, WhyNdayeni, Services, BusinessSolutions, CarePlans, Testimonials, TrustSignals, About, Contact, FAQ, Footer)
/services	Server	Services index page (9 services grouped by category)
/services/it-support-outsourcing	Server	IT Support service landing page
/services/computer-repairs	Server	Computer Repairs service landing page
/services/networking-wifi	Server	Networking & Wi-Fi service landing page
/services/cctv-security	Server	CCTV & Security service landing page
/services/printer-office-technology	Server	Printer & Office Tech service landing page
/services/web-design	Server	Web Design service landing page
/services/graphic-design-branding	Server	Graphic Design service landing page
/services/digital-automation	Server	Digital Automation service landing page
/services/digital-skills-training	Server	Digital Skills Training service landing page
/vacancies	Server	Public vacancies listing (empty state or cards)
/vacancies/[id]	Server	Vacancy detail page (full description, apply via email)
/vacancies/admin	Client	Standalone Vacancies admin (login + CRUD)
/privacy-policy	Server	POPIA-aligned privacy policy (14 sections)

Route	Type	Description
/terms	Server	Terms of Service (14 sections, SA legal references)
/training	Server	Academy landing page
/training/apply	Client	Academy application form
/training/forgot-password	Server	Two-mode: request reset link / set new password
/training/verify/[certificateNumber]	Client	Public certificate verification page

Admin Pages (2 routes)

Route	Type	Description
/training/admin	Client	SMS Admin Dashboard (11 views: Dashboard, Applications, Students, Courses, Attendance, Assessments, Certificates, Users, Reports, Audit Log, Settings)
/vacancies/admin	Client	Vacancies Admin (standalone, shared SSO with SMS)

6. API Endpoints

SMS API (/api/academy/)

Method	Route	Auth	Description
POST	/login	Public	Authenticate with email+password, sets session cookie
POST	/logout	Public	Clears session cookie
POST	/apply	Public	Submit academy application (rate-limited: 3/hour/IP)
GET	/students	Authenticated	List students (search, filter by status/course)
GET	/students?id=X	Authenticated	Single student profile (with relations + computed fields)
POST	/students	super/admin/admissions/training	Create/update/delete/convert student
GET	/courses	Public	List all courses with modules
POST	/courses	super/admin	Create/update/delete courses + modules
GET	/attendance	Authenticated	List attendance (filter by student/date)
POST	/attendance	super/admin/training	Create/update single OR bulk attendance
GET	/assessments	Authenticated	List assessments (filter by student OR course for gradebook)
POST	/assessments	super/admin/training	Create/update assessment
GET	/certificate	Public	View certificate by ID or certificateNumber
POST	/certificate	super/admin/admissions	Issue/manual-issue/reissue certificate
GET	/audit	Authenticated	Paginated audit log (filter by action/studentId/userId)
GET	/reports	Authenticated	Dashboard stats + 12-month trends + status breakdown + top courses
GET	/users	Authenticated	List all academy users
POST	/users	super (create) / self (edit/reset)	Create/update/resetPassword user

Method	Route	Auth	Description
DELETE	/users?id=X	super	Delete user (cannot delete self or last super)
GET	/export?type=X	Authenticated	CSV export (students/attendance/certificates)
GET	/welcome-letter?studentId=X	Authenticated	Download welcome letter PDF

Auth API (/api/academy/auth/)

Method	Route	Auth	Description
POST	/forgot-password	Public	Request password reset email (rate-limited: 3/hour/IP)
POST	/reset-password	Public	Reset password with token (rate-limited: 5/hour/IP)

Vacancies API (/api/academy/vacancies)

Method	Route	Auth	Description
GET	/vacancies	Public	List active vacancies (auto-hides expired, rate-limited: 60/min/IP)
GET	/vacancies?id=X	Public	Single vacancy detail (404 if draft/expired)
GET	/vacancies?admin=true	super/admin/admissions/training	All vacancies (incl. drafts + expired)
POST	/vacancies	super/admin/admissions/training	Create vacancy
GET	/vacancies/[id]	Public (active only) / Admin (all)	Single vacancy by ID
PUT	/vacancies/[id]	super/admin/admissions/training	Update vacancy (partial update)
DELETE	/vacancies/[id]	super/admin/admissions/training	Delete vacancy

Contact API

Method	Route	Auth	Description
POST	/contact	Public	Contact form submission (rate-limited: 5/hour/IP, SMTP email to info@ndayenisolutions.co.za)

7. Components

Public Site Components (`src/components/ndayeni/`)

Component	Description
<code>Navbar.tsx</code>	Fixed top nav with 3 dropdowns (Services, Training, Vacancies) + mobile burger menu with collapsible accordions
<code>Hero.tsx</code>	Hero section with headline, CTAs, stats, “What We Handle” panel, particle network background
<code>ParticleBackground.tsx</code>	Canvas 2D particle network (55 particles desktop, 30 tablet, 0 mobile)
<code>SectionDivider.tsx</code>	Gradient divider line between sections (server component, no JS)
<code>WhyNdayeni.tsx</code>	6 value-prop cards with GSAP entrance animation
<code>Services.tsx</code>	9 services in 3 category accordions with image headers
<code>BusinessSolutions.tsx</code>	3 solution cards (Business, Home, Ongoing Support)
<code>CarePlans.tsx</code>	3 IT support plan cards (Basic, Business featured, Business Plus)
<code>Testimonials.tsx</code>	Google Reviews CTA card + 4 trust badges
<code>TrustSignals.tsx</code>	4 compliance badges (POPIA, Secure by Default, SLA-Backed, Verified)
<code>About.tsx</code>	Founder story + 4-step process + stats + values
<code>Contact.tsx</code>	Smart contact form (customer type + interest checkboxes + reCAPTCHA-ready)
<code>FAQ.tsx</code>	10 questions in 3 categories (shadcn Accordion + JSON-LD FAQPage schema)
<code>Footer.tsx</code>	Quick Links + Services + Company Info + Service Area + Legal links + scroll-to-top
<code>WhatsAppButton.tsx</code>	Floating WhatsApp contact button

Admin/SMS Components (`src/components/academy/`)

Component	Description
<code>StudentProfileModal.tsx</code>	8-tab student detail modal (Overview, Personal, Education, Course, Attendance, Assessments, Certificates, Audit Trail)
<code>WelcomeLetterButton.tsx</code>	Download button for welcome letter PDF (locked until enrolled)
<code>AttendanceBulkCapture.tsx</code>	Bulk attendance: course picker + date + per-student radio (Present/Absent/Excused)
<code>AssessmentGradebook.tsx</code>	Gradebook matrix: students × modules with clickable cells + summary
<code>UserManagementPanel.tsx</code>	User CRUD: table + Add/Edit/Reset Password/Delete dialogs with role gating
<code>AuditLogViewer.tsx</code>	Filterable paginated audit log table with prefix-colored badges
<code>CSVExportButtons.tsx</code>	3 download links for students/attendance/certificates CSV
<code>ReportsCharts.tsx</code>	4 KPI cards + 4 recharts (Line, Bar, Bar, Pie donut)
<code>ForgotPasswordForm.tsx</code>	Two-mode: request reset link / set new password
<code>VacancyManagementPanel.tsx</code>	Vacancy CRUD: table + create/edit dialog + delete confirm + status toggle
<code>ServicePageTemplate.tsx</code>	Shared template for the 9 service landing pages

shadcn/ui Primitives (`src/components/ui/`)

44 pre-built components including: button, card, dialog, alert-dialog, table, tabs, accordion, select, input, textarea, badge, tooltip, dropdown-menu, popover, toast, toaster, and more.

8. Security Implementation

South African Compliance

- **POPIA** (Protection of Personal Information Act 4 of 2013): Privacy Policy page, data minimisation, audit trail, access controls, session security
- **Cybercrimes Act 19 of 2020**: Rate limiting, audit logging, security headers
- **ECT Act 25 of 2002**: HTTPS enforcement, HSTS, CSP
- **Consumer Protection Act 68 of 2008**: Fair Terms of Service, no fake testimonials

Authentication & Sessions

Feature	Implementation
Password hashing	PBKDF2 (10,000 iterations, 16-byte salt, 64-byte key, SHA-512)
Password validation	≥8 chars + uppercase + lowercase + digit + special char
Session tokens	HMAC-SHA256 signed (base64url payload + signature)
Session cookie	HttpOnly + Secure + SameSite=Strict + 24-hour expiry
Secret	<code>ACADEMY_SECRET</code> env var (mandatory, fail-fast if missing or < 16 chars)
Password verification	Constant-time comparison (timing-attack resistant)
Hash function	Async <code>pbkdf2</code> (non-blocking, no event-loop freeze)

Rate Limiting

In-memory sliding-window rate limiter (`src/lib/rate-limit.ts`):

Endpoint	Limit	Window
Login	5 attempts	15 minutes per IP
Login (per email)	5 failed attempts	15 minutes (account lockout)
Forgot password	3 requests	1 hour per IP
Reset password	5 attempts	1 hour per IP
Apply form	3 submissions	1 hour per IP
Contact form	5 messages	1 hour per IP
Vacancies (public read)	60 reads	1 minute per IP

HTTP Security Headers (via `next.config.ts`)

Header	Value
Content-Security-Policy	strict (default-src 'self', frame-ancestors 'none', object-src 'none')
X-Frame-Options	DENY
X-Content-Type-Options	nosniff
Referrer-Policy	strict-origin-when-cross-origin
Permissions-Policy	camera/microphone/geolocation/payment/usb disabled
Strict-Transport-Security	max-age=63072000; includeSubDomains; preload
X-DNS-Prefetch-Control	off
Cache-Control (API)	no-store, no-cache, must-revalidate

Audit Logging

Every admin action is logged to the `AuditLog` table: - `login.success` , `login.failed` , `login.locked_out` , `login.inactive_account` - `student.create` , `student.update` , `student.delete` , `student.status_change` , `student.convert` - `certificate.issue` , `certificate.manual_issue` , `certificate.reissue` - `attendance.bulk` - `user.create` , `user.update` , `user.reset_password` , `user.delete` - `vacancy.create` , `vacancy.update` , `vacancy.delete` - `welcome_letter.download` - `user.password_reset_requested` , `user.password_reset`

Input Sanitisation

- All user inputs sanitised server-side: trim, strip control chars, cap length
- HTML escaping for email content
- Prisma parameterised queries (SQL injection protected)
- React escapes on render (XSS protected)
- Employment type + status validated against allow-lists

9. Environment Variables

Required on Vercel (Production + Preview + Development):

Variable	Purpose	Example
DATABASE_URL	Supabase pooler connection (PostgreSQL)	postgresql://postgres.xxx:pass@aws-0-region.pooler.supabase.com:6543/postgres?pgbouncer=true
DIRECT_URL	Supabase direct connection (for migrations)	postgresql://postgres.xxx:pass@aws-0-region.supabase.com:5432/postgres
ACADEMY_SECRET	Session token signing secret (≥ 16 chars, use openssl rand -hex 32)	(random 64-char hex string)
SMTP_HOST	Mail server hostname	mail.ndayenisolutions.co.za
SMTP_PORT	Mail server port	587
SMTP_USER	SMTP username	info@ndayenisolutions.co.za
SMTP_PASSWORD	SMTP password	(mail account password)
CONTACT_TO_EMAIL	Where contact form submissions are sent	info@ndayenisolutions.co.za
NEXT_PUBLIC_SITE_URL	Canonical site URL	https://ndayenisolutions.co.za

Optional:

Variable	Purpose
NEXT_PUBLIC_CLARITY_ID	Microsoft Clarity project ID (heatmaps + session recordings)
NEXT_PUBLIC_RECAPTCHA_SITE_KEY	Google reCAPTCHA v3 site key (contact form)
RECAPTCHA_SECRET_KEY	Google reCAPTCHA v3 secret key (server-side verification)
SMTP_DISABLE_TEST_MODE	Set to "true" to disable Ethereum test email fallback

10. Performance Optimisations

iOS Safari Performance

Optimisation	Impact
<code>content-visibility: auto</code> on below-the-fold sections	Saves ~2.5MB initial image download (only above-the-fold images load on first paint)
<code>backdrop-filter: blur()</code> scoped to desktop only (<code>@media (hover: hover)</code>)	Eliminates 25 blur shader computations per scroll frame on touch devices
framer-motion removed entirely (SectionDivider rewritten as server component)	Saves ~50-100KB of JS bundle
Three.js HeroScene removed + replaced with Canvas 2D particle network	Saves ~600KB of JS (three.js + drei)
Images compressed to WebP q88	75% smaller than original JPGs (6.7MB → 1.7MB)
Particle background disabled on mobile (< 768px)	No Canvas animation on iPhone
GSAP scroll-linked parallax removed from Hero	Eliminates scroll repaints

Page Weight

Resource	Compressed Size
HTML	~32KB (Brotli)
JavaScript (14 chunks)	~294KB (Brotli)
CSS (2 files)	~30KB (Brotli)
Images on first load	~500KB (only above-the-fold, rest lazy-loaded via content-visibility)
Total first load	~856KB

11. SEO Infrastructure

Feature	Implementation
Sitemap	<code>src/app/sitemap.ts</code> — 16 entries (homepage, services index, 9 service pages, privacy-policy, terms, training routes)
robots.txt	<code>public/robots.txt</code> — allows all, disallows <code>/api/</code>
Metadata	Per-page <code>metadata</code> exports (title, description, canonical, OpenGraph, Twitter cards)
JSON-LD	<code>ProfessionalService</code> schema + 9 <code>Service</code> schemas in <code>layout.tsx</code>
OpenGraph image	<code>public/og-image.png</code> — 1200×630 PNG
PWA manifest	<code>public/manifest.json</code> — name, <code>theme_color</code> , icons
Semantic HTML	<code><main></code> , <code><header></code> , <code><nav></code> , <code><section></code> , <code><article></code> , <code><footer></code> throughout
Accessibility	Skip-to-content link, ARIA labels, keyboard navigation, sr-only classes

12. Analytics

Service	Status	Cookies?
Vercel Web Analytics	✅ Active (via <code>@vercel/analytics</code>)	No (privacy-friendly)
Microsoft Clarity	⚙️ Wired but inactive (<code>NEXT_PUBLIC_CLARITY_ID</code> not set)	Would use cookies if enabled
Google Analytics	❌ Not installed	—
reCAPTCHA v3	⚙️ Built into contact form but inactive (<code>NEXT_PUBLIC_RECAPTCHA_SITE_KEY</code> not set)	Would use cookies if enabled

Cookie consent banner: NOT needed — site uses no non-essential cookies.

13. Deployment

Hosting

- **Platform:** Vercel (auto-deploy on GitHub push to `main` branch)

- **Build command:** `prisma generate && next build && cp -r .next/static .next/standalone/.next/ && cp -r public .next/standalone/`
- **Output:** Standalone Next.js server (`output: "standalone"`)
- **Custom domain:** `ndayenisolutions.co.za` (apex → www redirect, Let's Encrypt SSL auto-managed by Vercel)

DNS (Afrihost)

Record	Type	Value
@ (apex)	A	Vercel IP (216.198.79.1)
www	CNAME	c5bd9362db9defd2.vercel-dns-017.com
_vercel	TXT	vc-domain-verify=ndayenisolutions.co.za,...
_vercel.www	TXT	vc-domain-verify=www.ndayenisolutions.co.za,...
MX	MX	mx6912926826.spe.ucebox.co.za (Afrihost email)
SPF	TXT	v=spf1 include:spf.aserv.co.za +a +mx -all
autoconfig	CNAME	envoy.aserv.co.za (email autodiscovery)
autodiscover	CNAME	envoy.aserv.co.za (email autodiscovery)

Database Setup

The `supabase-setup.sql` file contains the full database schema creation script. Run it in the Supabase SQL Editor when setting up a new database. It includes: - AcademyUser, Student, Course, Module, Attendance, Assessment, Certificate, AuditLog, Vacancy tables - Auto-update triggers for `updatedAt` fields - Seed data for 4 default courses + their modules

14. Development

Prerequisites

- Node.js 18+ or Bun 1.x
- A Supabase project (or local PostgreSQL)
- An SMTP mail server

Local Setup

```
# Clone the repository
git clone https://github.com/ndayenisolutions-png/Ndayeni-Solutions.git
cd Ndayeni-Solutions

# Install dependencies
bun install

# Set up environment variables
cp .env.example .env # then edit .env with your values

# Generate Prisma client
bun run db:generate

# Push database schema
bun run db:push

# Start dev server
bun run dev
```

Development Commands

Command	Description
<code>bun run dev</code>	Start Next.js dev server on port 3000
<code>bun run lint</code>	Run ESLint
<code>bun run build</code>	Production build (prisma generate + next build)
<code>bun run db:push</code>	Push Prisma schema to database (–accept-data-loss)
<code>bun run db:generate</code>	Regenerate Prisma client
<code>bun run db:migrate</code>	Create a Prisma migration
<code>bun run db:reset</code>	Reset database (destructive)

TypeScript Configuration

- **Strict mode:** enabled
- **Path aliases:** `@/*` → `./src/*`
- **Build errors:** `ignoreBuildErrors: false` (production builds fail on type errors)
- **Excluded directories:** `Ndayeni-Solutions/`, `examples/`, `skills/`, `mini-services/`, `tool-results/`, `tests/`

15. Admin Credentials

The admin account is stored in the `AcademyUser` table: - **Email:**

nhlakanipho@ndayenisolutions.co.za - **Role:** super - **Password:** (PBKDF2 hashed — not stored)

in plaintext)

Passwords are hashed using PBKDF2 with 10,000 iterations, 16-byte salt, 64-byte derived key, SHA-512 digest. The hash format is `salt:derivedHex`.

16. Git History Summary

Major Milestones (most recent first)

1. Welcome letter professional formatting (margins, fonts, spacing)
 2. Particle network background (Canvas 2D, 55 particles)
 3. Tablet dropdown fix (group-focus-within)
 4. iOS performance: content-visibility:auto + framer-motion removed
 5. Admin sidebar: position:fixed (never scrolls)
 6. Login text simplification
 7. Security hardening (rate limiting, headers, sessions, password complexity, audit logging)
 8. Solutions tab removed from navbar
 9. Vacancies dropdown in navbar
 10. Vacancies admin standalone (decoupled from SMS)
 11. Vacancies page + admin management
 12. SMS admin sidebar sticky fix
 13. Mobile burger menu: collapsible accordions
 14. Image re-encoding (WebP q88)
 15. Hero: 3D objects + background image removed
 16. iOS Safari performance (backdrop-filter scoped, three.js removed, images compressed)
 17. Final recommendations: legal pages, 9 service pages, FAQ, testimonials, analytics, manifest, sitemap
 18. SMS polish: student profile, bulk attendance, gradebook, cert reissue, reports trends, forgot/reset password, CSV export, welcome letter PDF
 19. Database: switch to PostgreSQL (Supabase)
 20. Initial project setup + marketing site build
-

17. Known Limitations & Future Work

Not Yet Implemented

- Blog / resources section (for content marketing + SEO)
- Case studies / portfolio page
- Team page
- Newsletter signup
- Google reviews integration on testimonials section
- Google Business Profile optimisation

- Custom 404 page
- Cookie consent banner (not needed currently — no non-essential cookies)
- Service area map (Google Maps embed)
- Live chat widget
- CV submission form (currently uses mailto: link)

Security Notes

- `ACADEMY_SECRET` is mandatory (fail-fast if missing) — must be set in Vercel env vars
- Session tokens expire after 24 hours (users must re-login daily)
- Rate limiting is in-memory (per Vercel instance, not shared across instances) — sufficient for current traffic volume
- reCAPTCHA is built into the contact form but inactive (rate limiting provides sufficient bot protection)
- No CSRF token (SameSite=Strict cookie provides CSRF defence)
- No 2FA (not implemented — recommended for future if admin team grows)

Performance Notes

- Particle network background is desktop/tablet only (disabled on mobile for iPhone performance)
- Below-the-fold sections use `content-visibility: auto` (images lazy-load on scroll)
- `backdrop-filter: blur()` is desktop-only (touch devices use solid rgba backgrounds)
- Images are WebP q88 (good quality, ~75% smaller than original JPGs)

End of Document